

ECON726: Policy and Program Evaluation

Tara Sullivan

Instructor

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Office hour: Tuesday, 5:30-6:30, 1542 West,
(and by appointment)

Course Information

Class Location: North Bldg C114
Time: Tuesday, 7:30-9:30 PM

Course Description

In real-world settings, analysts must draw inferences about causes and effects from observational, i.e., non-randomized, data. For example, how do we truly determine whether charter schools produce better student outcomes, or whether a marketing campaign has increased consumer awareness? This course will cover statistical methods for causal inference to help analysts design and analyze observational studies for real-world decision-making.

Prerequisites

- Statistical analysis software is the primary tool for this course, and I will be providing course instructions in python.
- You are required to have taken a semester-long course in probability and statistical inference.
- Ideally, you have completed coursework in regression methods; otherwise, the econometric theory and the technical approach for the course will be difficult to follow.
- This is a course in econometrics and applied statistics; as such, it is expected that you have a strong interest in business and public policy applications.

Grading Policy

Empirical project (80% of final grade)

The bulk of your grade will come from an empirical project. The project is based on an empirical paper of your choice, which (1) uses a method discussed in this course; (2) has data available online; (3) is not based on an experiment.

The paper should:

1. Be based on a paper published in a peer-reviewed journal.
2. Briefly state the research question and discuss the dataset.
3. Provide a (formal) discussion of the identification strategy (What is the ideal randomized experiment? why is it causal?).

4. Replicate (selected) main findings.
 - This should not be directly taken from the replication package online! I will check!
5. Describe and implement some sort of extension to this paper. This could be new and interesting tests for key assumptions, robustness checks, additional results based on other methods or data, or a combination thereof.

Paper selection: Due **September 28** [5% of final grade]

- Select your paper. Briefly describe the causal effect of interest and methodology. Discuss what you think you will do for your extension of this paper.
- If your paper is not appropriate, I'll let you know at this stage, and you will have to find another.

Project plan: Due **October 12** [15% of final grade]

- This paper should be your first effort at addressing points (1) and (2) above. Summarize the public policy or business context of the paper. Write down the specific question the paper is evaluating. Discuss the data you are using (both your replication data, and data for your extension). Verify that you can access the data and replication package. Discuss the extensions you are considering, and, if necessary, where you will be getting that data. Verify that you've read the paper

Replication results: Due **November 2:** [10% of final grade]

- You should submit some of your main replication results. The final results should be formatted as tables and charts in a word document or PDF, submitted on Brightspace. Your code and data must be accessible on GitHub.

Presentation: Due **December 1 & December 8** [10% of final grade]

- Prepare a 5-8 minute presentation on your paper. Focus on the research question, main result, and one extension.

Final Paper: Due **December 11** [40% of final grade]

- Your paper should contain 6-10 pages of text, and 3-6 pages of figures (double spaced, 12 point font). You must provide the code and data to replicate your findings. Your final project should be submitted on Brightspace, and your code and data should be uploaded to github (note: if your data is too large to upload to github, let me know).

Notes:

- Deadlines are midnight the night before class
- Please let me know **within the first two weeks** if you need to adjust any deadlines (check you other course deadlines!)
- *Late policy:* This project is cumulative, and split into separate assignments.
 - For each assignment, 10% is deducted for each day the item is late.
 - If an assignment is missed completely, partial credit (up to 50%) can be earned by **successfully** completing the final submission. For example, if you miss selecting a paper, you can earn 50% of the "Paper selection" grade by handing in your final submission.

Homework (10% of final grade)

Early in the course, short homeworks will occasionally be assigned. **These homeworks will be graded for completion. Please DO NOT use AI to complete these assignments; there is no advantage in doing so.** These assignments will largely consist of coding exercises and simple mathematical derivations. The goal is to ensure you have a working understanding of the tools taught so far, and are not falling behind.

Additionally, readings will be assigned throughout the course. Please complete appropriate readings prior to class. Short pop quizzes based on the readings may be assigned, depending on overall participation.

Attendance and Participation (10% of final grade)

This course does not have a virtual component; attendance is expected.

AI Policy

This course is meant to extend your theoretical and empirical abilities. Over-reliance on LLMs will hamper that effort. Overall, my aim is to reward genuine effort, over seemingly perfect code and text. To that end, I encourage you to avoid AI for your work. I understand it may be helpful for brainstorming and troubleshooting, but please ensure that your written code and words are your own.

That said, **unthinking AI usage will be heavily penalized**. If I can tell that you thoughtlessly used an AI, that will result in a zero on your assignment. Note that chat bots often produce lackluster essays that sound acceptable to the untrained. Those who heavily used AI tend to do poorly on the final assignment, as they generally miss the mark. Those who pay attention in class and follow the recommended steps tend to do well.

To be transparent about my AI usage: I will primarily use LLMs to brainstorm, troubleshoot, and check course materials. I've been advised by other professors to run your code through AI for grading purposes (as a secondary grader); if you would prefer I don't do that, please don't hesitate to let me know.

Overall, this policy is meant to be a dialogue, not dogma. Please feel free to let me know your thoughts.

Required textbook

The primary textbook for this course is **Mostly Harmless Econometrics** (Angrist and Pischke 2009). This textbook is meant to provide a readable introduction to many topics of this course. I did not list it on the Hunter College bookstore, because it is much cheaper if you order on Amazon ([link](#))

Additional readings will be assigned throughout the semester, and are from free resources.

Additional resources

These are additional resources I used in developing the course.

Causal Inference for the Brave and True (Alves 2022)

- Excellent online companion to Mostly Harmless; I will cite from this website throughout the course. Effectively brings the methods from Mostly Harmless into the machine learning age, with sample code in Python. Be aware there are (very occasional!) errors.

Applied Causal Inference Powered by ML and AI (Chernozhukov et al. 2025)

- Covers similar topics as Mostly Harmless, but from an ML perspective. Currently available free online.

Additional resources may prove helpful if you choose to dive deeper into these topics. Please note that none of the textbooks below are required.

Business Data Science (Taddy 2019)

- Sometimes called the “machine learning Mostly Harmless.” I will borrow heavily from Chapters 5 (Experiments) and 6 (Controls). Note that the code provided in the book is written in R.

Causal Inference for Statistics, Social, and Biomedical Sciences: An Introduction (Imbens and Rubin 2015)

- A more technical introduction to causal models, largely taught through the lens of the potential outcomes model. If you want to better understand the derivations in this course, or delve into details on standard errors, you should consult this resource.

Causal Inference: The Mixtape (Cunningham 2021)

- <https://mixtape.scunning.com/>

Course outline

1. Introduction/idealized experiment
2. Probability review
3. Statistics review
4. Potential outcomes and experiments
5. Linear regression models
6. Matching and Propensity models
7. Difference-in-differences models
8. Synthetic Control
9. Machine learning methods in Causal inference

Note the following dates when class is cancelled:

- October 13 (classes follow Monday schedule)

References

- Alves, Matheus Facure. 2022. “Causal Inference for the Brave and True.” Accessed August 18, 2025. <https://matheusfacure.github.io/python-causality-handbook/landing-page.html>.
- Angrist, Joshua D., and Jörn-Steffan Pischke. 2009. *Mostly Harmless Econometrics: an Empiricist’s Companion*. Princeton, NJ: Princeton University Press.
- Casella, George, and Roger L. Berger. 2002. *Statistical Inference*. Second Edition. Belmont, CA: Brooks/Cole Cengage Learning.
- Chernozhukov, Victor, Christian Hansen, Nathan Kallus, Martin Spindler, and Vasilis Syrgkanis. 2025. “Applied Causal Inference Powered by ML and AI.” Accessed August 18, 2025. <https://causalml-book.org/>.
- Cunningham, Scott. 2021. “Causal Inference: The Mixtape.” Accessed August 22, 2025. <https://mixtape.scunning.com/>.
- Imbens, Guido W., and Donald B. Rubin. 2015. *Causal Inference for Statistics, Social, and Biomedical Sciences: An Introduction*. New York, NY: Cambridge University Press.
- Taddy, Matt. 2019. *Business Data Science*. New York, NY: McGraw-Hill Education.